

Zeros of Linear Functions from Tables

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A **linear function** is a function whose graph is a straight line.

Zero of a Function

The **zero of a function** is the value of x that makes the function value equal to zero.

Zero in a Table

In a table of values, the **zero** occurs when $y = 0$.

Finding the Zero

To determine the zero of a linear function from a table, locate the row where the output value is 0.

When $y = 0$ Is Not Listed

If $y = 0$ does not appear directly in the table, use the pattern of change in the values to determine the corresponding x -value.

Zero and x -Intercept

The zero of a linear function corresponds to the x -**intercept** of its graph.

Examples

Example 1

Determine the zero of the linear function represented in the table.

x	-2	-1	0	1	2
y	-6	-3	0	3	6

Example 2

Determine the zero of the linear function represented in the table.

x	0	1	2	3	4
y	8	6	4	2	0

Example 3

Determine the zero of the linear function represented in the table.

x	-3	-2	-1	0	1
y	4	2	0	-2	-4

Practice Exercises

Practice 1

Determine the zero of the linear function represented in the table.

x	-2	-1	0	1	2
y	-6	-3	0	3	6

Practice 2

Determine the zero of the linear function represented in the table.

x	0	1	2	3	4
y	8	6	4	2	0

Practice 3

Determine the zero of the linear function represented in the table.

x	-3	-2	-1	0	1
y	4	2	0	-2	-4

Practice 4

Determine the zero of the linear function represented in the table.

x	1	2	3	4	5
y	-4	-2	0	2	4

Practice 5

Determine the zero of the linear function represented in the table.

x	-1	0	1	2	3
y	6	3	0	-3	-6

Practice 6

Determine the zero of the linear function represented in the table.

x	-4	-3	-2	-1	0
y	12	9	6	3	0

Practice 7

Determine the zero of the linear function represented in the table.

x	2	3	4	5	6
y	-6	-3	0	3	6

Practice 8

Determine the zero of the linear function represented in the table.

x	-5	-4	-3	-2	-1
y	-4	-2	0	2	4

Practice 9

Determine the zero of the linear function represented in the table.

x	0	2	4	6	8
y	10	5	0	-5	-10

Practice 10

Determine the zero of the linear function represented in the table.

x	-2	0	2	4	6
y	6	3	0	-3	-6

Activities

Activity 1

Determine the zero of the linear function represented in the table.

x	-3	-2	-1	0	1
y	-8	-4	0	4	8

Activity 2

Determine the zero of the linear function represented in the table.

x	1	2	3	4	5
y	12	9	6	3	0

Activity 3

Determine the zero of the linear function represented in the table.

x	-2	-1	0	1	2
y	6	3	0	-3	-6

Activity 4

Determine the zero of the linear function represented in the table.

x	0	1	2	3	4
y	-6	-3	0	3	6

Activity 5

Determine the zero of the linear function represented in the table.

x	-4	-3	-2	-1	0
y	8	6	4	2	0

Activity 6

Determine the zero of the linear function represented in the table.

x	3	4	5	6	7
y	-6	-4	-2	0	2

Activity 7

Determine the zero of the linear function represented in the table.

x	-1	0	1	2	3
y	-8	-4	0	4	8

Activity 8

Determine the zero of the linear function represented in the table.

x	-6	-5	-4	-3	-2
y	-6	-3	0	3	6

Activity 9

Determine the zero of the linear function represented in the table.

x	1	3	5	7	9
y	12	6	0	-6	-12

Activity 10

Determine the zero of the linear function represented in the table.

x	-4	-2	0	2	4
y	8	4	0	-4	-8

Thank you!